



Nagar Yuwak Shikshan Sanstha's
Yeshwantrao Chavan College of Engineering
(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)
Hingna Road, Wanadongri, Nagpur - 441 110



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Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Artificial Intelligence and Data Science

Vision of the Department

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission of the Department

To nurture computer engineering professionals by offering:

- Dynamic learning environment that encourages problem-solving using emerging technology
- National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration
- Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	Mission: Means to achieve Vision
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Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Krishna Mishra

(Signature in Handwritten)



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Session	2026-27 (Odd)	Course Name	LAB: Optimization technique
Semester	7th	Course Code	23ADS1708
Roll No	41	Name of Student	Krishna Mishra

Practical Number	8
Course Outcome	CO1: Understand unconstrained and constrained optimization techniques. CO2: Apply non-linear optimization techniques to solve engineering problems. CO3: Analyze optimization problems using operations research techniques. CO4: Evaluate transportation and transshipment optimization problems.
Aim	To determine the shortest paths using the Floyd–Warshall Algorithm.
Problem Definition	To find the shortest distance between every pair of vertices in a weighted directed graph with 5 vertices using the Floyd–Warshall Algorithm. The edges (with weights) are: 1→2 (3), 1→3 (8), 1→5 (7), 2→4 (1), 3→2 (4), 4→1 (2), 4→3 (5), 5→4 (6).
Theory (100 words)	The Floyd–Warshall algorithm is a dynamic programming method that finds the shortest paths between all pairs of vertices in a weighted graph. It works with negative edge weights but not with negative cycles. Let $d[i][j]$ be the shortest known distance from vertex i to vertex j . The algorithm takes each vertex k in turn as a possible intermediate vertex and updates $d[i][j] = \min(d[i][j], d[i][k] + d[k][j])$. After all n vertices have been considered, d holds the shortest distances. The time complexity is $O(n^3)$ and the space complexity is $O(n^2)$. A negative value on the diagonal indicates a negative cycle.



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	Step-by-Step Method 1. Represent the graph as an $n \times n$ matrix: 0 on the diagonal, the edge weight where an edge exists, and ∞ where there is no edge. 2. Initialize the distance matrix D with this matrix. 3. For $k = 1$ to n, take vertex k as the intermediate vertex. 4. For every pair (i, j), update $d[i][j] = \min(d[i][j], d[i][k] + d[k][j])$. 5. After all n iterations, the matrix contains the shortest distance between every pair of vertices. 6. Check the diagonal of the matrix for negative values to detect a negative cycle. 7. Reconstruct the actual shortest paths using the next-vertex matrix.
Procedure and Execution (100 Words)	Algorithm: 1. Define the weighted graph as an adjacency matrix. 2. Initialize the distance matrix and the next-vertex matrix. 3. For each intermediate vertex k, for each pair (i, j): 4. if $d[i][k] + d[k][j] < d[i][j]$, update $d[i][j]$ and $next[i][j]$. 5. Check the diagonal for a negative cycle. 6. Display the final shortest distance matrix. 7. Reconstruct and display the shortest path for every pair of vertices. Code: <pre>INF = float("inf") # Weighted directed graph (adjacency matrix), vertices 1 to 5 graph = [[0, 3, 8, INF, 7], [INF, 0, INF, 1, INF], [INF, 4, 0, INF, INF], [2, INF, 5, 0, INF], [INF, INF, INF, 6, 0]] n = len(graph) def show(matrix, title): print(title) print(" " + "".join(f"{j+1:>5}" for j in range(n))) for i in range(n): row = "".join(f"{'INF' if v == INF else int(v):>5}" for v in matrix[i])</pre>



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```
        print(f"{i+1:>4} {row}")
    print()

# Distance and next-vertex matrices
dist = [row[:] for row in graph]
nxt = [[j if graph[i][j] != INF and i != j else None for j in range(n)]
        for i in range(n)]

show(dist, "Initial Distance Matrix:")

# Floyd-Warshall algorithm
for k in range(n):
    for i in range(n):
        for j in range(n):
            if dist[i][k] + dist[k][j] < dist[i][j]:
                dist[i][j] = dist[i][k] + dist[k][j]
                nxt[i][j] = nxt[i][k]

show(dist, "Final Shortest Distance Matrix:")

# Negative cycle check
if any(dist[i][i] < 0 for i in range(n)):
    print("Graph contains a negative cycle")

# Path reconstruction
def get_path(u, v):
    if nxt[u][v] is None:
        return None
    path = [u + 1]
    while u != v:
        u = nxt[u][v]
        path.append(u + 1)
    return path

print("Shortest Paths:")
for i in range(n):
    for j in range(n):
        if i != j:
            p = get_path(i, j)
            if p is None:
                print(f"{i+1} -> {j+1}: no path")
            else:
                route = " -> ".join(map(str, p))
                print(f"{i+1} -> {j+1}: cost = {int(dist[i][j])}, path = {route}")
```

Output:

Initial Distance Matrix:

	1	2	3	4	5
1	0	3	8	INF	7
2	INF	0	INF	1	INF
3	INF	4	0	INF	INF
4	2	INF	5	0	INF
5	INF	INF	INF	6	0

Final Shortest Distance Matrix:

	1	2	3	4	5
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	<pre> 1 0 3 8 4 7 2 3 0 6 1 10 3 7 4 0 5 14 4 2 5 5 0 9 5 8 11 11 6 0 Shortest Paths: 1 -> 2: cost = 3, path = 1 -> 2 1 -> 3: cost = 8, path = 1 -> 3 1 -> 4: cost = 4, path = 1 -> 2 -> 4 1 -> 5: cost = 7, path = 1 -> 5 2 -> 1: cost = 3, path = 2 -> 4 -> 1 2 -> 3: cost = 6, path = 2 -> 4 -> 3 2 -> 4: cost = 1, path = 2 -> 4 2 -> 5: cost = 10, path = 2 -> 4 -> 1 -> 5 3 -> 1: cost = 7, path = 3 -> 2 -> 4 -> 1 3 -> 2: cost = 4, path = 3 -> 2 3 -> 4: cost = 5, path = 3 -> 2 -> 4 3 -> 5: cost = 14, path = 3 -> 2 -> 4 -> 1 -> 5 4 -> 1: cost = 2, path = 4 -> 1 4 -> 2: cost = 5, path = 4 -> 1 -> 2 4 -> 3: cost = 5, path = 4 -> 3 4 -> 5: cost = 9, path = 4 -> 1 -> 5 5 -> 1: cost = 8, path = 5 -> 4 -> 1 5 -> 2: cost = 11, path = 5 -> 4 -> 1 -> 2 5 -> 3: cost = 11, path = 5 -> 4 -> 3 5 -> 4: cost = 6, path = 5 -> 4 </pre>
Output Analysis	The program computed the shortest distance between all pairs of vertices. Distances that were initially infinite became finite through intermediate vertices. For example, $d(1, 4) = 4$ through $1 \rightarrow 2 \rightarrow 4$ ($3 + 1$), $d(5, 2) = 11$ through $5 \rightarrow 4 \rightarrow 1 \rightarrow 2$ ($6 + 2 + 3$), and $d(3, 5) = 14$ through $3 \rightarrow 2 \rightarrow 4 \rightarrow 1 \rightarrow 5$ ($4 + 1 + 2 + 7$). All diagonal values are 0, so the graph has no negative cycle.
Link of student Github profile where lab assignment has been uploaded	https://github.com/krish-na45/Optimization-Technique-Practicals
Conclusion	The shortest paths between all pairs of vertices were successfully determined using the Floyd–Warshall Algorithm. The final distance matrix and the corresponding shortest routes were obtained for the given 5-vertex graph.



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